

$$9) \ y = (5x^4 - 3x^2 - 1)(-5x^2 + 3)$$

$$10) \ f(x) = (-10x^2 - 7\sqrt[5]{x^2} + 9)(2x^3 + 4)$$

$$11) \ y = (5 + 3x^{-2})(4x^5 + 6x^3 + 10)$$

$$12) \ y = (-6x^4 + 2 + 6x^{-4})(6x^4 + 7)$$

$$13) \ f(x) = \left(-7x^4 + 10x^{\frac{2}{5}} + 8\right)(x^2 + 10)$$

Critical thinking question:

- 14) A classmate claims that $(f \cdot g)' = f' \cdot g'$ for any functions f and g . Show an example that proves your classmate wrong.